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Techno-economic optimization and environmental Life Cycle Assessment (LCA) of microgrids located in the US using genetic algorithm

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ABSTRACT

A methodology was developed that assessed the techno-economic and environmental performance of a small scale microgrid located in US cities of Tucson, Lubbock and Dickinson. Providing uninterrupted power the microgrid was composed of seven components - solar photovoltaics, wind-turbines, lead acid batteries, biodiesel generators, fuel cells, electrolyzers and H_2 tanks. Firstly, detailed mathematical models that predicted the hourly energy generation for each of the components were developed and validated. Secondly, based on an electricity dispatch strategy, configurations having lowest LCOE (Levelized Cost of Energy) were determined using an evolutionary optimization technique - Genetic Algorithm (GA). Results for a single home microgrid were verified using an exhaustive search technique that scanned the entire design space to find the lowest LCOE configuration. The microgrid size was subsequently increased to satisfy power requirements for 10 and 50 homes and new lowest LCOE configurations were determined using GA to examine the economies of scale effect on sustainability. The LCOEs obtained were in the range of $0.32–0.42/kWh and were also compared with similar economic analyses available in the literature. The carbon footprint (LCA GHG emissions – CO_2 eq.) was extremely low and was approximately 1/10th as that of an equivalent conventional electric grid.

1. Introduction

Today's world remains heavily reliant on fossil fuel sources to meet its energy demand. In 2015, approximately 80% of the total demand was met by combusting fossil fuels. Furthermore, relative to 1973 levels, the carbon dioxide (CO_2) emissions attributed to fossil fuel combustion has doubled to about 32,294 metric tons [1]. To decrease the reliance on fossil fuels and mitigate CO_2 emissions, transition can be made towards power generation using more renewable resources. The potential for this energy transition seems likely based on estimates that by 2022, power generation via renewables will increase globally by over 33% with solar and wind resources together expected to account for more than 80% of the total generating capacity growth [1].

A major challenge in widespread implementation of renewable resources is their natural intermittency. To mitigate the impact of this problem, hybrid energy systems that combine various renewable resources in a 'microgrid' have been considered. Previous studies show this approach can significantly increase overall system efficiency and reliability [2]. Microgrids are defined as a group of interconnected loads and electricity generation units which also connect to the main grid but can operate in standalone mode [2]. Researchers have shown microgrids to be more economical than systems which rely on a single renewable energy resource [3]. For remote locations such as islands and off-grid communities, microgrids have proven economically superior to a conventional electric grid [4]. However, selection of the most efficient combination of renewable resources in a microgrid remains a crucial issue for reducing the impact of intermittency while ensuring optimal economic performance. Recent studies have focused on finding the optimum microgrid size and design when considering regional annual electricity load profiles and the availability of renewable resources. The optimization sizing problem is both non-convex and non-linear, so multiple solutions are possible due to the sheer number of input variables (capacities of solar PV, wind turbines, batteries, etc.) [5]. As a result, designing an optimal microgrid can be computationally intensive and time consuming. Therefore, stochastic solution techniques such as evolutionary algorithms have been used to tackle this problem. Maleki et al. adopted an artificial bee swarm optimization (ABSO) algorithm to design and size a lowest cost microgrid composed of solar photovoltaic panels (PV), wind turbine (WT) and fuel cell (FC) systems [5]. Shang et al. used an improved particle swarm optimization (PSO) technique to size batteries for a hybrid energy system composed of PV, WT and diesel generator (DG) systems. A cost minimizing objective function was employed to analyze scenarios with different penetration levels of renewable energy in a microgrid [6]. For a remote area in Iran, Maleki

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Nomenclature

List of abbreviations

ABSO	artificial bee swarm optimization
AZ	Arizona state
BDG	Biodiesel Generators
BP4175B	PV solar panel assumed in this study
C_{batt} (t)	instantaneous battery capacity at the end of time instant 't' (Ah)
C_{BD}	biodiesel consumption of BDG (lit/h)
C_{eff} (t)	charging efficiency at the end of time instant 't'
C_k	component (k) price which includes the acquisition and installation costs
$CMtn_k$	maintenance costs of component 'k' ($)
C_{nom}	nominal values of charge capacity (Ah)
CO_2	carbon dioxide
C_p	specific heat of H_2 (kJ/kg/K)
CRF	capital recovery factor (%)
d	discount factor (%)
DER-CAM	Distributed Energy Resources Customer Adoption model
DG	Diesel Generators
E_{act}	activation overpotential for FC (V)
E_{cell}	cell voltage for FC (V)
E_{con}	concentration overpotential for FC (V)
ED (t)	energy deficit available at time 't'
EE (t)	excess energy available at time 't'
e_k	emission factor for component 'k' (kg CO_2 eq emitted/kWh generated)
EM	electrolyzer module
EMS	Energy Management Strategy (EMS)
E_{oc}	open circuit voltage for FC (V)
E_{ohm}	Ohmic overpotential for FC (V)
ERIBI (t)	energy residing in battery at beginning of instant't' (kWh)
ESA	exhaustive search algorithm
ETR	extraterrestrial radiation
N_b	number of battery units
ND	North Dakota state
N_{H2}	H_2 produced by EM (moles per hour)
N_k	number of units of a single type of component 'k'
$N_{k,max}$	upper bound integer value for component 'k'
NREL	National Renewable Energy Laboratory
P_1	compressor's inlet pressure (MPa)
P_2	compressor's outlet pressure (MPa)
P_{EM}	power consumption of the EM (W)
PEMFC	proton exchange membrane fuel cell
P_{gen}	energy generated by BDG (kWh)
P_{gen} (t)	hourly power generated by microgrid (kWh) at time instant 't'
P_k	present values (per unit) of installation and acquisition costs for component 'k'
p_k	annual power generated by the component 'k' (kWh)
P_L (t)	residential hourly load (kWh) at time instant 't'
$P_{L,an}$	annual electricity demand (kWh)
P_{NG}	nominal power rating of BDG (kWh)
P_R	rated power capacity of wind turbine (kW)
PSO	particle swarm optimization
PV	Solar Photovoltaics
PVC	present value of costs for acquisition, installation, maintenance and replacement ($)
r	isentropic exponent of H_2
SAM	System Advisor Model
SID	Supplementary Information Document
T	Microgrid life (yrs)
T_1	compressor's inlet temperature (K)
TAC	total annualized cost ($)
T_{amb} (t)	ambient temperature at time instant 't' (°C)
TE	total annual emissions (kg CO_2 eq emitted/year)
TMY3	typical meteorological year data set
f	diffuse fraction
F	Faradays constant (C)
FC	Fuel Cell
f_k	cloudy conditions factor
GA	Genetic Algorithm
$G_{b,\beta}$	hourly beam (or direct) solar radiation (W/m^2)
$G_{b,\beta}$	tilted beam component (W/m^2)
$G_{d,\beta}$	sky diffuse component of solar radiation (W/m^2)
G_{dh}	horizontal diffuse component of solar radiation (W/m^2)
GHG	greenhouse gases
GHI	global horizontal irradiance
g_k	annual fall in acquisition costs for component 'k' (%)
$G_{r,\beta}$	reflected component of solar radiation (W/m^2)
H	hub height (m)
H_2	hydrogen
HOMER	Hybrid Optimization of Multiple Energy Resources
I	current density (A/cm^2)
i	annual inflation rate (%)
I_{batt} (t)	instantaneous current at end of time instant 't' (A)
I_{nom}	nominal values of current (A)
$Inst_k$	installation costs of component 'k'
ISOC (t)	instantaneous state of charge at the end of time instant 't' (Ah)
k	clearness index
LB	Lead Acid battery
LCA	Life Cycle Assessment
LCOE	levelized cost of energy
Lg_k	acquisition costs' drop limiting value (%)
LPSP	loss of power supply probability
m_c	mass flow rate of H_2 entering the compressor (kg/s)
Mtn_k	maintenance costs of component 'k' as a % of initial costs (P_k)
TX	Texas state
v	wind speed (m/s)
V_{batt} (t)	instantaneous battery voltage at the end of time instant 't' (V)
v_C	cut-in wind speed (m/s)
V_{elec}	operating voltage of EM (V)
v_F	cut-off speed (m/s)
V_i	voltage efficiency of EM (%)
V_{nom}	nominal values of voltage (V)
v_R	rated wind speed (m/s)
W_c	compressor's power requirement (W)
WT	Wind Turbine
Y_{mat}	time required to reach limiting value of acquisition costs' drop – Technology Maturation period (yrs)

List of Greek symbols

α	wind speed power law coefficient
β	PV panel tilt (°)
θ	incidence angle (°)
θ_z	Zenith angle (°)
η_c	compressor efficiency (%)
η_i	current efficiency of electrolyzer (EM)
ρ	albedo

et al. designed a PV, WT and battery-based hybrid energy system to provide the lowest electricity cost using a PSO based Monte Carlo method [7]. A similar techno-economic study for four different regions in Iran was conducted in which a hybrid energy system of PV, WT, FC and battery banks were optimally sized. The hybrid systems were found to have lower costs than those systems with a single renewable energy component [8]. Ramli et al used a self-adaptive differential evolution algorithm with multiple optimization objectives including cost minimization, lost power supply probability (LPSP) and maximum renewable energy fraction to design a hybrid energy system in Yanbu, Saudi Arabia. This system was composed of PV, WT, batteries and DGs [9]. Using HOMER software, Phurailatpam et al. sized a microgrid for rural and urban applications in India with the lowest net present cost and minimum CO_2 emissions [10]. To size a microgrid, another study considered a triple multi-objective optimization analysis using a genetic algorithm to minimize cost, CO_2 emissions and unmet load [11]. Other studies used multi-objective optimization techniques to size a microgrid while simultaneously minimizing LCOE and CO_2 emissions [12,13]. Somma et al. [14] and Rezvani et al. [15] used a multi-objective optimization method to consider economic and environmental aspects for operation scheduling in microgrids. A similar study was done by Elsied et al. who used a binary particle swarm multi-objective optimization method to minimize cost and emissions [16].

Previously mentioned studies were mainly focused on finding the optimal microgrid size with the lowest levelized cost of energy (LCOE) and LPSP value [5–9]. Some studies also evaluated the microgrid's carbon footprint [10–16]. Researchers generally used commercial software tools such as HOMER (Hybrid Optimization of Multiple Energy Resources), DER-CAM (Distributed Energy Resources Customer Adoption Model) and TRNSYS to perform techno-economic analyses and environmental assessments based on location, load and available renewable energy resources [10,17]. These tools allow a quick microgrid assessment but restrict the user from seeing the actual equations being solved (black-box operation). The ability to view, modify or add more sophisticated mathematical models for various renewable energy technologies is restricted in these "black-box" commercial tools. Considering that the entire microgrid model is only as accurate as its individual component submodels, the individual components must be validated against experimental data or accepted software tools to ensure accuracy. Unfortunately, none of the above-mentioned studies provided any validation of the reported results using these "black-box" software tools. Also, comparison of final results with similar microgrid analyses was not reported in the previous studies nor was any comparison with those of an equivalent conventional electric grid [7–17]. Since a microgrid must compete with the existing conventional electric grid to justify its construction and use, the techno-economic and environmental results from a microgrid analysis must be compared to a conventional electric grid. Also, previous studies have not evaluated the effect of microgrid size on LCOE and CO_2 emissions. The present work addresses these existing research gaps and provides four novel contributions including:

- Development of techno-economic and environmental assessment models for a microgrid composed of PV, WT, LB, FC, BDG, EM and H_2 tanks with submodel validation of PV, WT, and FC by direct comparison to experimental data or accepted software tools.
- Optimal microgrid sizing for lowest cost using a GA with solution verification using an exhaustive search (brute force) method for a single home system in 3 US cities with significantly different renewable energy resource availability.
- Microgrid scaling to 10 and 50 homes to examine "economy of scale" effect on LCOE ($/kWh) and CO_2 emissions.
- Direct comparison of LCOE and CO_2 emissions from a microgrid to an equivalent conventional electric grid and to similar studies available in the literature to assess the potential for adoption of a microgrid in terms of economic feasibility and environmental sustainability.

The methodology followed in this work considered three U.S. residential communities, namely, Tucson, Arizona, Lubbock, Texas and Dickinson, North Dakota. A schematic diagram of the general microgrid configuration is shown in Fig. 1. The technical, economic and environmental assessment metrics considered in this study included LPSP, LCOE ($/kWh) and CO_2 emissions (kg eq. emitted/year). This work was carried out in three steps. During the first step, electricity generated by primary sources such as PV and WT was calculated for each location and individual module generating capacity. In the second step, based on hourly load and electricity generated, an electricity dispatch strategy was adopted that estimated the power generated by secondary sources including LB, BDG, FC and EM. In the third step, using a sizing algorithm, an optimum microgrid configuration was identified that provided uninterrupted power at the lowest annual LCOE. The optimal solution was identified using a genetic optimization algorithm (GA) which was then verified by an exhaustive search algorithm (ESA) for a single home system. As simulations became more computationally intensive and time consuming (for 10 and 50 homes), only the GA was

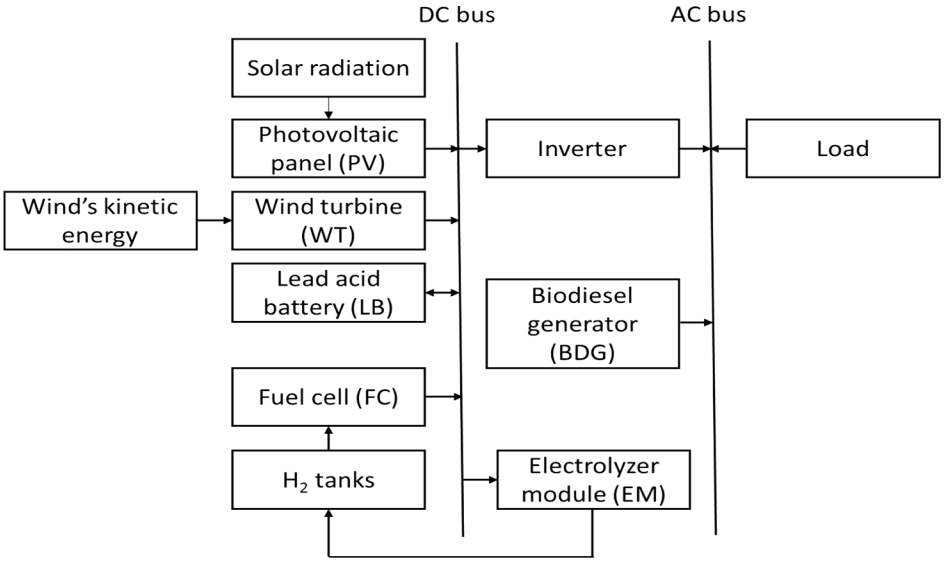


Fig. 1. Schematic diagram of proposed microgrid configuration.

used to identify the optimum microgrid configuration due to its accuracy and rapid execution time. The LCOE and CO_2 emissions of the electricity generated by the microgrid were then compared with those from a conventional electric grid.

This paper starts with Section 2 that describes the renewable energy potential for the cities considered followed by Section 3 that presents details of the mathematical submodels for each individual microgrid component and formulates the optimization problem. Section 4 deals with the optimization algorithm utilized to find the "best" microgrid configuration and Sections 5 and 6 present results and conclusions respectively.

2. Renewable energy potential of considered cities

The solar and wind energy potentials of the three US cities are depicted in Fig. 2. The hourly solar radiation (direct normal irradiance) averaged for the entire year in case of Tucson is 0.29 kWh/m^2. This value is 20% and 64% higher than that of Lubbock and Dickinson respectively. However, the wind potential, in terms of yearly averaged wind speed (m/s) for Tucson is 3.58 m/s it is 59% and 44% less than of Lubbock and Dickinson respectively. These values indicate that, amongst all cities, Dickinson has highest wind potential but lowest solar potential while Tucson has highest solar potential but lowest wind potential. The solar and wind potential of Lubbock lies in between the other two cities. The location specific hourly values for solar radiation and wind speed were procured from typical meteorological year (TMY3) data set which is available online at National Renewable Energy Laboratory's (NREL) website [18].

3. Model development

This section deals with the mathematical modeling of individual components in a microgrid from systems perspective. The technical model is presented first, followed by economic and environmental emissions' models. Information regarding the validation of energy generation models with experimental data or relevant software tools is provided in a supplementary information document (SID) available on request.

3.1. Technical model development

The following section describes individual component models including PV, WT, LB, BDG, FC, EM and H_2 tanks.

3.1.1. Solar photovoltaic model

The solar photovoltaic model is divided into two parts, namely, solar radiation model and PV panel model.

The solar radiation model shown here estimates the amount of solar radiation incident on a tilted solar panel, G_β (W/m^2) and given by [19]:

$$G_\beta(t) = G_{b,\beta}(t) + G_{r,\beta}(t) + G_{d,\beta}(t) \quad (1)$$

Here, $G_{b,\beta}$, $G_{r,\beta}$, $G_{d,\beta}$ are the hourly beam (or direct), reflected and the sky diffuse components of solar radiation respectively. To calculate these components, measured hourly solar radiation data for extra-terrestrial radiation on a horizontal surface (ETR) and the global horizontal irradiance (GHI) were procured from the TMY3 data set [18].

To calculate $G_{b,\beta}$, the value of clearness index (k) was computed using [20]:

$$k(t) = \frac{GHI(t)}{ETR(t)} \quad (2)$$

The Diffuse fraction (f) was calculated using the following correlations [20]:

$$f(t) = 0.995 - 0.081k(t) for k(t) \leq 0.21 \quad (3)$$

$$f(t) = 0.724 + 2.738k(t) - 8.321k^2(t) + 4.967k^3(t) \ for \ 0.21 < k(t) \leq 0.76 \quad (4)$$

$$f(t) = 0.18 \ for \ k(t) > 0.76 \quad (5)$$

The value of the horizontal diffuse component (G_{dh}) was computed from the following equation [20]:

$$G_{dh}(t) = f(t)GHI(t) \quad (6)$$

Finally, $G_{b,\beta}$ is calculated using the Klucher model [21]:

$$G_{b,\beta}(t) = (GHI(t) - G_{dh}(t))\frac{\cos\theta(t)}{\cos\theta_z(t)} \quad (7)$$

Here θ and θ_z are incidence and zenith angles respectively. The equations to compute their hourly values are provided in the SID.

The Reflected beam component ($G_{r,\beta}$) was estimated assuming that the diffuse and reflected components of solar radiation were isotropic in nature thus $G_{r,\beta}$ can be computed as [21]:

$$G_{r,\beta}(t) = \frac{\rho}{2}GHI(t)(1 - \cos\beta) \quad (8)$$

Here β is the tilt of the solar panel (taken as 20°) and ρ is the albedo (taken as 0.2).

The Tilted diffuse component ($G_{d,\beta}$) is estimated by incorporating a factor 'f_k', which takes into account the effect of cloudy conditions on solar irradiance and calculated by [21]:

$$f_k(t) = 1 - \left(\frac{G_{dh}(t)}{GHI(t)}\right)^2 \quad (9)$$

With this, the final expression for $G_{d,\beta}$ is given as [21]:

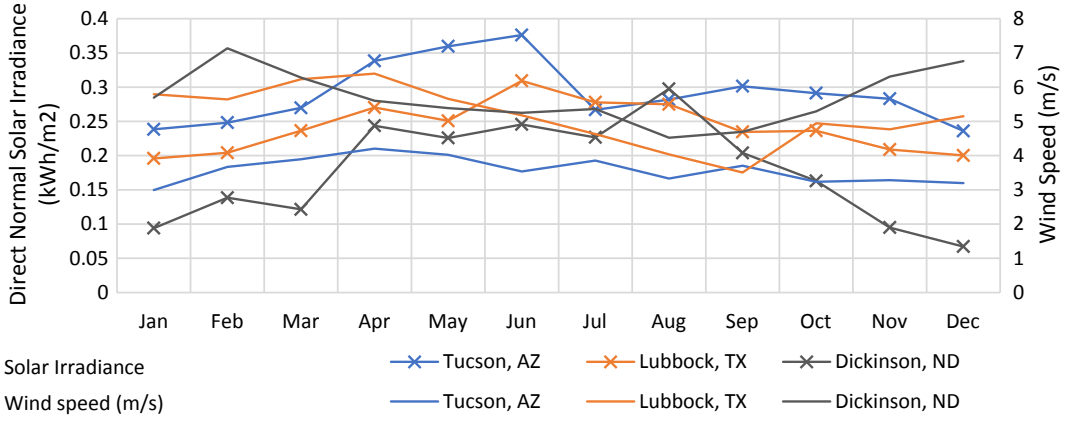


Fig. 2. Solar and wind energy potential for all the three US cities.

$$G_{d,\beta}(t) = G_{dh}(t)\left[0.5\left(1+\cos\frac{\beta}{2}\right)\right]\left[1+f_k(t)\left(\sin\frac{\beta}{2}\right)^3\right][1 + f_k(t)(\cos\theta(t))^2(\sin\theta_z(t))^3] \quad (10)$$

Which provides an estimate for G_β in Eq. (1) using estimated values of $G_{b,\beta}$, $G_{r,\beta}$ and $G_{d,\beta}$ from Eqs. (7), (8) and (10) respectively.

For all the three cities, the monthly G_β values of solar radiation using the above were validated with the corresponding values estimated by National Renewable Laboratory's (NREL) software tool known as System Advisor Model (SAM) [22]. The information regarding the validation of these results are provided in the SID.

The PV panel model predicts the hourly power (DC) generated primarily based on values of G_β. The PV module was assumed to always operate at maximum power with the hourly maximum power current (I_{mp}) and voltage (V_{mp}) found by simultaneously solving the two non-linear equations as follows [23]:

$$\frac{I_{mp}(t)}{V_{mp}(t)} = \left[\frac{\frac{I_o(t)}{a(t)}exp\left(\frac{V_{mp}(t)+I_{mp}(t)R_s(t)}{a(t)}\right)+\frac{1}{R_{sh}(t)}}{1+\frac{R_s(t)}{R_{sh}(t)}+\frac{I_o(t)R_s(t)}{a(t)}exp\left(\frac{V_{mp}(t)+I_{mp}(t)R_s(t)}{a(t)}\right)}\right] \quad (11)$$

$$I_{mp}(t) = I_L(t) - I_o(t)\left[exp\left(\frac{V_{mp}(t)+I_{mp}(t)R_s(t)}{a(t)}\right)-1\right] - \left[\frac{V_{mp}(t)+I_{mp}(t)R_s(t)}{R_{sh}(t)}\right] \quad (12)$$

To solve Eqs. (11) and (12), five parameters are required namely, light current (I_L), diode reverse saturation current (I_o), modified ideality factor (a), series resistance (R_s) and shunt resistance (R_{sh}). Hourly values of these parameters were calculated using their own reference condition values ($T_{c,ref}$ = 25 °C and $G_{\beta,ref}$ = 1000 W/m^2). For the BP4175B module (power capacity 175 W), reference parameter values for $I_{L,ref}$, $I_{o,ref}$, a_{ref}, $R_{s,ref}$, $R_{sh,ref}$ were taken as 5.467A, 1.452 × 10^{-9} A, 1.982 V, 0.495 Ω and 155 Ω, respectively [22]. Expressions to compute hourly values of these five parameters are provided in the following sections.

3.1.2. Module (cell) temperature (T_c)

The hourly value of T_c (°C) depends on two factors, namely, the temperature of the surroundings (T_{amb}, °C) and the amount of incident solar radiation, G_β (W/m^2) and was given by [23]:

$$T_c(t) = T_{amb}(t) + (T_{NOCT}-20)\frac{G_\beta(t)}{800}\left(1-\frac{\eta_c}{0.9}\right) \quad (13)$$

Here, T_{amb} was obtained from TMY3 file [20], T_{NOCT} (46.5 °C) is the nominal operating cell temperature, and η_c is the module or cell efficiency (13.9%) [22,23].

3.1.3. Light current (I_L)

I_L (A) depends linearly on G_β and T_c and was calculated by [23]:

$$I_L(t) = \frac{G_\beta(t)}{G_{\beta,ref}}[I_{L,ref} + \mu_{I,sc}(T_c(t) - T_{c,ref})] \quad (14)$$

Here, $\mu_{I,sc}$ (0.005 A/°C) was taken as the short circuit temperature coefficient for the solar module [22].

3.1.4. Modified ideality factor (a)

a (V) depends linearly on the cell temperature (expressed in K) and was computed by [23]:

$$\frac{a}{a_{ref}} = \frac{T_c}{T_{c,ref}} \quad (15)$$

3.1.5. Shunt resistance (R_{sh})

R_{sh} (Ω) was assumed to be dependent on G_β and was estimated by [23]:

$$\frac{R_{sh}(t)}{R_{sh,ref}} = \frac{G_{\beta,ref}}{G_\beta(t)} \quad (16)$$

3.1.6. Reverse diode saturation current (I_o)

I_o (A) was assumed to be dependent on T_c, material bandgap for Silicon (E_g, J) and was computed by [23]:

$$\frac{I_o(t)}{I_{o,ref}} = \left(\frac{T_c(t)}{T_{c,ref}}\right)^3 exp\left[\frac{1}{k}\left(\frac{E_{g,ref}}{T_{c,ref}} - \frac{E_g(t)}{T_c(t)}\right)\right] \quad (17)$$

Here, k was the Boltzmann constant (1.38 × 10^{-23} J/K). The value of E_g was estimated by [23]:

$$\frac{E_g(t)}{E_{g,ref}} = 1 - 0.000267(T_c(t) - T_{c,ref}) \quad (18)$$

The value for $E_{g,ref}$ was taken as 1.794 × 10^{-19} J [23].

Using Eqs. (13)–(18), hourly values for I_L, a, R_{sh} I_o were estimated and were plugged into Eqs. (11) and (12) to find I_{mp} and V_{mp}. Subsequently, hourly power, $P_{gen,PV}$ (W) produced by the module was computed by [23]:

$$P_{gen,PV}(t) = V_{mp}(t)I_{mp}(t) \quad (19)$$

Above equation computes the energy generated by a single PV module of 175 W capacity. 5.7 modules were connected in series to increase the capacity to 1 kW. The monthly power generated by these modules was validated with the corresponding values estimated by SAM software [22] and these results are provided in the SID.

3.1.7. Wind turbine (WT) model

The instantaneous power generated by the wind turbine depends on three types of wind velocities namely, the cut-in wind speed v_C (m/s), the rated wind speed v_R (m/s) and the cut-off speed v_F (m/s). These wind speeds in turn depend on the type and model of wind turbine deployed. There are numerous models that are available in the literature that predict power output of a wind turbine. Tina et al. used a linear model [24], while Diaf et al. [25] and Mohammadi et al. [26] utilized a quadratic and cubic model respectively. In this work, a quadratic equation was employed as it was shown to better predict values when validated with values calculated by SAM software. For WT model, a Bergey BWC XL wind turbine of 1 kW rated capacity (P_R) was chosen and the values assumed for v_C, v_R and v_F were 3 m/s, 12 m/s and 23 m/s respectively [22]. Based on these wind speed values, the hourly power generated by the wind turbine, $P_{gen,WT}$ (kW), as a function of wind speed (v) was calculated using the following equation [25]:

$$P_{gen,WT}(t) = \begin{cases} P_R\left(\frac{v^2(t)-v_c^2(t)}{v_R^2(t)-v_c^2(t)}\right) for v_c \le v(t) \le v_R \\ P_R \;\; for \;\; v_R < v(t) < v_F \\ 0 \;\; for \;\; v(t) < v_c \;\; and \;\; v(t) > v_F \end{cases} \quad (20)$$

Location specific hourly wind speed data at 100 m was procured from NREL's database [27]. The corresponding wind speeds (v), adjusted to 30 m hub height (H) were calculated using the following equation [25]:

$$v(t) = v_{ref}(t)\left(\frac{H}{H_{ref}}\right)^\alpha \quad (21)$$

Here α is the wind speed power law coefficient and its value can be taken as "1/7" assuming well exposed sites [25]. The power generated by wind turbine was computed using the above methodology and was validated with the values generated by SAM software and its results are provided in the SID.

3.1.8. Lead acid battery (LB) model

Electrochemical energy storage through lead acid batteries are widely used in microgrids due to their low capital and maintenance costs as compared to Lithium ion batteries [28]. In this work, a normalized and generalized LB model was used as it had shown excellent ability to represent a wide range of lead acid batteries [29,30].

In the studies that focus on microgrid analyses, most of them employed a simple energy balance to compute the energy stored in the battery and its state of charge [3,4,6,11]. The simplified battery model used in these studies did not take into account the transient behavior of battery variables which is depicted by current, voltage, capacity and charging efficiency. Therefore in this work, transient modeling of such variables was carried out and their values were computed for every hour in a year.

For simulation purposes, the nominal values of voltage (V_{nom}), current (I_{nom}) and capacity (C_{nom}, at 10 h discharge rate) for a single electrochemical accumulator were assumed to be 2 V, 55A and 550 Ah respectively [29,30]. A single battery unit was assumed to be composed of 20 such accumulator connected in series amounting to a total energy capacity of 22 kWh.

3.1.9. Instantaneous state of charge (ISOC)

The state of a charge is a measure of energy stored in the battery. Its value is not only dependent on the energy residing in battery at beginning of instant't' (ERIBI (t)), but also on the values of instantaneous current (I_{batt}), capacity (C_{batt}) and the voltage (V_{batt}) of the battery at the end of time instant 't'. The expressions for ISOC (shown below) were based on the coulomb counting technique described by Gergaud et al. [29] and Guasch et al [30]. During charging regime, the value for ISOC at the end of time instant 't' was given by [29,30]:

$$ISOC(t) = \frac{ERIBI(t) + I_{batt}(t)V_{batt}(t)C_{eff}(t)}{C_{batt}(t)V_{batt}(t)} \tag{22}$$

During discharging regime, the value for ISOC is given by [29,30]:

$$ISOC(t) = \frac{ERIBI(t) - I_{batt}(t)V_{batt}(t)}{C_{batt}(t)V_{batt}(t)} \tag{23}$$

3.1.10. Instantaneous battery capacity (C_{batt})

C_{batt} denotes the amount of energy (Ah) that the battery can restore. Its transient behavior depends on the amount of current flowing through it (I_{batt} (t)), ambient temperature condition (T_{amb}) and was expressed by the following equation [29,30]:

$$C_{batt}(t) = \frac{1.67C_{nom}[1 + 0.005(T_{amb}(t) - 25)]}{\left[1 + 0.67\left(\frac{|I_{batt}(t)|}{I_{nom}}\right)^{0.9}\right]} \tag{24}$$

3.1.11. Charging and discharging efficiency

A coulombic charging and discharging efficiency model is adopted in this work which neglects any losses associated due to battery's internal resistances. The battery's discharging efficiency was assumed to be 100%. However, C_{eff} is a function of ISOC and I_{batt} and was given by [29,30]:

$$C_{eff}(t) = 1 - exp\left[\frac{20.73(ISOC(t) - 1)}{\frac{I_{batt}(t)}{I_{nom}} + 0.55}\right] \tag{25}$$

3.1.12. Instantaneous battery voltage (V_{batt})

V_{batt} depends on its internal elements, namely, the electromotive force and the internal resistance and is also affected by T_{amb}. V_{batt} during charging regime of battery was given by [29,30]:

$$V_{batt}(t) = N_b[2 + 0.16ISOC(t)] + N_b\frac{I_{batt}(t)}{C_{nom}}\left[\frac{6}{1 + I_{batt}(t)^{0.86}} + \frac{0.48}{(1 - ISOC(t))^{1.2}} + 0.036\right](1 - 0.025(T_{amb}(t) - 25)) \tag{26}$$

V_{batt} during discharging regime was expressed by the following equation [29,30]:

$$V_{batt}(t) = N_b[1.965 + 0.12ISOC(t)] - N_b\frac{|I_{batt}(t)|}{C_{nom}}\left[\frac{4}{1 + |I_{batt}(t)|^{1.3}} + \frac{0.27}{ISOC(t)^{1.5}} + 0.02\right](1 - 0.007(T_{amb}(t) - 25)) \tag{27}$$

In the above equations, N_b is the number of battery units. The sequence of calculation of all the above battery variables is provided in the SID.

3.1.13. Biodiesel powered electric generator (BDG)

Microgrids have been frequently designed with the inclusion of a conventional diesel powered generator [3,4,9,11,13]. Few techno-economic studies incorporate a biodiesel powered generator [10,12]. On a life cycle basis, biodiesel emits 66–72% less greenhouse gas (GHGs) emissions than conventional diesel [31]. Therefore, in this work, with an objective to enhance sustainability and reduce the overall carbon footprint, a biodiesel powered generator was chosen to provide electric power to the microgrid. To generate the same amount of power in an electric generator, the specific fuel consumption (kg/kWh) of biodiesel is higher than that of conventional diesel and the relative increase in fuel consumption varies between 4.6% and 11.9% [32]. Assuming an average value of 8%, the biodiesel consumption, C_{BD} (lit/h) of the electric generator was given by [11]:

$$C_{BD} = 1.08(0.08145P_{NG} + 0.2476P_{gen}) \tag{28}$$

Here P_{NG} (kWh) is the nominal power rating of the generator and P_{gen} (kWh) is the power generated by it.

3.1.14. Fuel cell (FC) model

Fuel cells (FC) are characterized by their ability to generate clean power at low operating temperatures [33]. Proton exchange membrane fuel cells (PEMFC), a type of FCs, have been widely used for power generation purposes in automobiles as well as in stationary and portable electricity generation units. They are widely used due to their capability of generating power with efficiencies of 40–60% which are higher than most of the other energy conversion pathways [33].

A PEMFC consists of a cathode and an anode separated by a polymer electrolyte membrane. Pure H_2 is fed to anode where it is oxidized, while oxygen (mostly via air) is supplied to cathode where it is reduced, subsequently producing water and heat.

The voltage (E_{cell}) of the FC was computed using the following expression [34]:

$$E_{cell} = E_{oc} - E_{act} - E_{con} - E_{ohm} \tag{29}$$

where E_{oc}, E_{act}, E_{con}, E_{ohm} denote the values for open circuit voltage, activation overpotential, concentration overpotential and ohmic overpotential respectively. E_{cell} was calculated by implementing the one dimensional mathematical model developed by Abdin et al. [34] in Matlab. For modeling a single PEMFC, an active area of 51.84 cm^2 [34] and an electrical efficiency of 42% was assumed [35]. The model consisted of four parts, namely, anode (9 equations), cathode (8 equations), membrane (8 equations) and voltage (22 equations). The complete mathematical model was sequentially solved. The information with regards to FC's mathematical model, its physical parameters and its validation with experimental data is provided in the SID.

By referring power-current density curve, the power generated by a single PEMFC was estimated to be approximately 25 W. Forty such PEMFCs (N_{FC}) were then assumed to be connected in series to scale up

the capacity to 1 kW. H_2 consumption (N_{H2}) in moles per hour by the entire unit was calculated using the following expression [34]:

$$N_{H2} = \frac{S_{H2} I N_{FCs}}{2F} \times 3600 \quad (30)$$

Here, the stoichiometry molar ratio of H_2 at anode (S_{H2}) and Faraday's constant was taken to be 1.2 and 96,485 C/mol respectively [34]. The value of current (I) was estimated to be 39.47 A. Inserting all these values in the Eq. (30), the H_2 consumption rate of a 1 kW PEMFC capacity was computed to be 35.34 mol/h. The systems level interaction and energy balance between H_2 cylinder and PEMFC is shown in Fig. 3.

3.1.15. Electrolyzer module (EM)

An electrolyzer is a FC operating in a reverse direction splitting water molecules into H_2 and O_2 by consuming electrical energy. It consists of two electrodes (cathode and anode) separated by a PEM [36]. Protons from anode diffuse through the PEM while electrons travel through the external circuit to reach cathode, where they combine together to produce H_2 gas. For modeling purposes, current (η_i) and voltage efficiencies (η_v) were assumed to be 100% and 74% respectively [37]. The operating voltage (V_{elec}) of the EM was calculated using the following equation [37]:

$$\eta_v = \left(\frac{1.48}{V_{elec}}\right) \times 100 \quad (31)$$

The H_2 produced in moles per hour (N_{H2}) by the EM was computed using the Faraday's law of electrolysis as follows [37]:

$$N_{H2} = \left(\frac{P_{EM}}{V_{elec} 2F}\right) \times 3600 \quad (32)$$

where P_{EM} (W) is power consumption of the EM. For an EM of 1 kW rated capacity, N_{H2} was determined to be 9.33 mol per hour. The discharge pressure of the EM was 0.6 MPa based on experimental studies reported in the literature [37].

3.1.16. Compressor and H_2 cylinders

The H_2 discharged from the EM was pumped to 13.5 MPa through a compressor and was stored in cylinder. The power requirement (W_c) of a compressor was computed by [37]:

$$W_c = \frac{C_p T_1 m_c}{\eta_c}\left[\left(\frac{P_2}{P_1}\right)^{\frac{r-1}{r}} - 1\right] \quad (33)$$

The values for all the parameters in the above equation are provided in the SID. The total energy consumed by the EM and compressor to inject 9.33 mol of H_2 in the cylinder was 1.045 kWh. The systems level energy balance and the interaction between the EM, compressor and the H_2 cylinder is shown in Fig. 4.

3.1.17. Electricity consumption model

For all the cities, the average hourly electricity usage for a single home was procured from NREL's database [38]. For a single home, the average hourly electricity load for the all the three cities are depicted in Fig. 5. The usage for 10 and 50 homes were linearly extrapolated by assuming that every home in a specific location used equal amounts of electricity.

3.2. Economic and environmental assessment

Prior studies on Microgrid have included the minimization of CO_2 emissions in their main optimization objectives [11–16]. Even though the minimization of carbon emissions is a vital objective, however, the only way to accelerate the widespread public acceptance and implementation of microgrids is to devise solutions that possess the least cost. An environmentally friendly solution would be difficult to finance if it is an expensive one. Therefore, in this work, the sole objective was to determine microgrid configurations which possess the lowest cost of energy. The carbon footprint of these configurations was later assessed.

3.2.1. Objective function

The expression to calculate LCOE ($/kWh), and the objective (minimization) function was given by [9,37]:

$$LCOE = \frac{TAC}{P_{L,an}} \quad (34)$$

Here $P_{L,an}$ was the annual electricity demand (kWh). TAC ($) was the total annualized cost of the microgrid and was given by the following equation [25]:

$$TAC = PVCCRF \quad (35)$$

Here, PVC is the present value of costs for acquisition, installation, maintenance and replacement of microgrid components. CRF is the capital recovery factor and was expressed by [25]:

$$CRF = \frac{d(1+d)^T}{(1+d)^T - 1} \quad (36)$$

Here, d was the discount factor (8%) and T was the lifetime of microgrid (20 years).

3.2.2. Effect of inflation and technology maturation on costs

Prior microgrid analyses utilized simple cost estimation models to perform economic assessments. For instance, the effect of inflation on capital and maintenance costs was neglected in some analyses [4,5–10,12]. The typical lifetime of microgrid is 20–25 years and during that time some components (LB, FC, EM) may need replacement every 5 years. Therefore, neglecting the effect of inflation on capital cost of such components may result in an inaccurate cost estimation. Furthermore, these studies also neglected the fall in costs due to technology maturation. It was therefore assumed that in this work, the acquisition costs for these components fall 10% ($g_k = -0.1$) every year until they reach they reach 50% ($Lg_k = -0.5$) of their initial acquisition value (at the time of initial investment, t = 0 years). This time (Y_{mat}, in years) for reaching 50% reduction was calculated by [39]:

$$Y_{mat} = \frac{\log(1 + Lg_k)}{\log(1 + g_k)} \quad (37)$$

For components such as LB, FC and EM, the annual inflation (i) rate of 3% was applied only after the maturation period ended (Y_{mat} = 7 years). For inverters, the inflation rate was applicable from the beginning (time, t = 0).

3.2.3. Acquisition costs and installation costs

The present values, C_k (per unit) for every component (k) comprised of installation ($Inst_k$) and acquisition costs (P_k, shown in Table 1) were calculated based on the following equations [39]:

For PV, WT, BDG and H_2 the equations were given by [39]:

$$C_k = P_k(1 + Inst_k) \quad (38)$$

For LB, FC and EM components, equations were [39]:

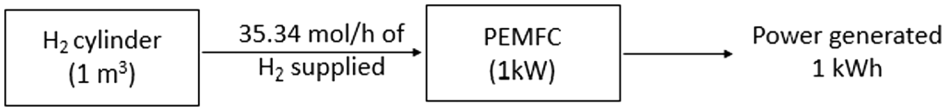


Fig. 3. Systems level interaction and energy balance between H2 cylinder and PEMFC.

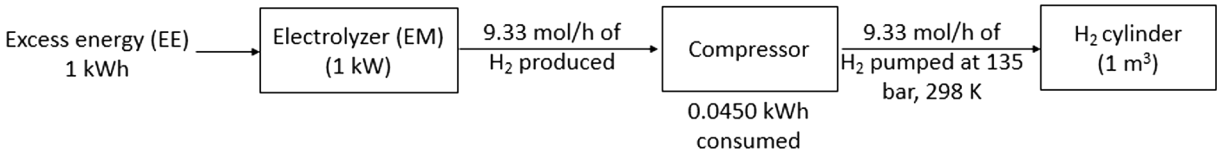


Fig. 4. Systems level interaction and energy balance between electrolyzer (EM), compressor and hydrogen cylinder.

$$C_k = P_k(1 + Inst_k)\left[1 + \left(\frac{1+g_k}{1+d}\right)^5 + \frac{(1+g_k)^7(1+i)^3}{(1+d)^{10}} + \frac{(1+g_k)^7(1+i)^8}{(1+d)^{15}}\right] \quad (39)$$

For inverter the costs were computed by [39]:

$$C_{Inv} = P_{Inv}(1 + Inst_{Inv})\left[1 + \left(\frac{1+i}{1+d}\right)^{10}\right] \quad (40)$$

The acquisition, installation and maintenance costs of components for a small scale microgrid (up to 50 homes) are shown in Table 1 and were assumed to be the same for all the cities. The biodiesel consumption costs were assumed to be $0.91/lit [40].

3.2.4. Maintenance costs

The maintenance costs ($CMtn_k$) were estimated as a direct percentage of initial cost of component (k) in its first year and can be given by [25,39]:

$$CMtn_k = \sum_{T=1}^{T=20} P_k Mtn_k \left(\frac{1+i}{1+d}\right)^T \quad (41)$$

Here, Mtn_k is the percentage of the component's acquisition cost and for all the components their respective values were shown in Table 1.

Finally, TAC for the entire microgrid was calculated by [25,39]:

$$TAC = CRF \sum_{k=1}^{k=7} (C_k + CMtn_k)N_k \quad (42)$$

Here N_k denotes the number of units of a single type of component (k).

3.2.5. Constraints

The seven variables (components) are integer constrained. All these variables are bounded as follows:

$$0 \le N_k \le N_{k,max} \quad (43)$$

Here N_k not only denotes the number of components of a specific component (k) but it also denotes the capacity of that component in kW. For instance, 5 PV components indicates that their net generation capacity is 5 kW. This is true for all components except LB and H_2 cylinder. Each LB denotes a storage capacity of 22 kW and each H_2 cylinder indicates a volumetric capacity of 1 m^3.

The value of $N_{k,max}$ was chosen in such a way that every power generating component (PV, WT, BDG, FC) can comfortably fulfil the peak electricity demand. The maximum value for battery capacity (22 kWh) was chosen in such a way that it can alone fulfil the mean power demand for 10–12 h. Similarly, the capacity of H_2 cylinder (1 m^3) was chosen so that it can meet power demand (via FCs) for 2–3 days.

The LB model employed in this work is only operational when the state of charge (ISOC) lies between 20% and 80% which can be given by:

$$0.2 \le ISOC(t) \le 0.8 \quad (44)$$

The probability of power supply failure (LPSP) was calculated at every hour in a year by using the following formula [5]:

$$LPSP = \sum_{t=1}^{8760} \left(\frac{P_L(t) - P_{gen}(t)}{P_L(t)}\right) \quad (45)$$

Here, P_L (t) and P_{gen} (t) are the residential load and power generated in the microgrid respectively.

The maximum and minimum pressure rating of the hydrogen cylinders were chosen to be 135 bar and 27 bar [45]. The cylinder model was only operational if the pressure lied between these limits.

$$27bar \le Pressure(t) \le 135bar \quad (46)$$

3.2.6. Environmental assessment

The GHG emissions (kg CO_2 equivalent) are estimated based on the amount of energy produced (or stored in case of batteries) by each component. A life cycle approach (LCA) was adopted to compute the emissions. The emissions (kg CO_2 eq./kWh generated) for every component are shown in Table 2.

The total annual emissions (TE) of the microgrid was computed using the following equation:

$$TE = \sum_{k=1}^{k=7} (p_k e_k) \quad (47)$$

Here p_i (kWh) is the annual power generated by the component 'k' and e_k is the amount of CO_2 emitted for every unit of electricity (kWh) generated.

4. Methodology

In this section, initially an energy management strategy (EMS) for

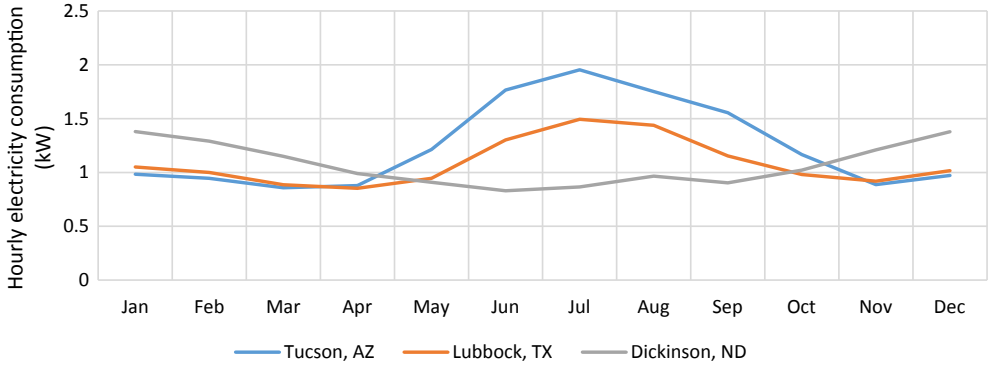


Fig. 5. Average hourly electricity load for a single home in all the three cities [35].

Table 1
Acquisition, installation, maintenance costs of components along with their life spans.

Module name	Acquisition costs, P_k ($/W)	Installation costs, $Inst_k$ (%)	Maintenance costs, Mtn_k (%)	Life span (yrs)
Solar PVs (PV) [41]	1.11	40	1	20
Wind turbines (WT) [41]	1.23	20	2	20
Lead acid battery (LB), $/Wh [41]	0.3	20	0	5
Biodiesel generator (BDG) [42]	0.84	10	2	20
Fuel cell (FC) [43]	5.19	10	2	5
Electrolyzer with compressor [5,11,37]	7.6	10	2	5
H_2 cylinder, $/cylinder [44]	300	10	0	20
Inverter (Inv) [41]	0.71	10	1	10

Table 2
LCA GHG emissions (kg CO_2 equivalents/kWh) [12].

Component	LCA GHG emissions (kg CO_2 eq./kWh), e_k
PV module (mono- Si)	0.045
WT (at speed 6.5 m/s)	0.011
Lead acid battery (per kWh stored)	0.028
Biodiesel generator	0.191
Fuel cell	0.664
Electrolyzer and H_2 tanks	0.011
US conventional electric grid [43]	0.51

the microgrid was proposed. The EMS was implemented through GA and ESA techniques. The simulations have been carried out using Matlab software and a computer having a dual core processor (i5-3230M) with 4 GB of memory.

4.1. Energy management strategy (EMS)

The hourly EMS of the microgrid was identical for both the algorithms and is depicted in Fig. 6. In the first step, based on meteorological data, electricity generation by PV and WT units (if available) was calculated for a time instant (t). In the next step, based on the difference between electricity generated and demand, either the amount of excess electricity (EE) or the deficit (ED) was calculated. At this point, the main EMS diverges into two paths. One for EE and the other for ED. In case of EE path, the battery (if available) model runs in a charging mode while in case of ED route, the battery runs in a discharging mode. After discharging mode ends, if more energy is needed (ED $>$ 0), then BDG and FC units (if available) are run in sequential order. After the LB has run in charging mode, if excess energy is still available (EE $>$ 0), then EM (if available) unit was operated to generate H_2 gas. This H_2 gas is then compressed and stored in cylinders. If storage space was not available then H_2 was not generated. Any EE available after this step was sold back to the conventional grid at wholesale rate (assumed to be 50% of retail price in respective states). It was also assumed that the infrastructure required for transmitting and distributing electricity back to the conventional grid already exists, and therefore costs related to that were neglected. The retail cost of electricity per kWh for Tucson, Lubbock and Dickinson were taken to be $0.1033, $0.0843 and $0.0894 respectively [44]. In the final step, LPSP was calculated.

4.2. Genetic Algorithm (GA)

A GA algorithm was utilized to find the optimum microgrid configuration able to provide uninterrupted power (LPSP = 0) at the lowest cost of energy (LCOE). The GA flowchart is depicted in Fig. 7.

Initially, as a first population, a random set of integer combinations (solutions), each one consisting of seven variables was generated. A solution (N_{PV} N_{WT} N_{LB} N_{BDG} N_{FC} N_{EM} N_{H2}) represented a single microgrid configuration. The LCOE (objective function) was computed after implementing the EMS for the entire year. At the end of any hour, if the value of LPSP was determined to be non-zero, then the implementation of EMS was stopped and a high numerical value is assigned to objective function LCOE ($100/kWh). This was done to steer the GA away from undesirable solutions (microgrid configurations that fail to provide continuous power). The value of LCOE is a measure of fitness of a solution in the GA. Lower the value of LCOE, higher is the fitness function value. After the LCOEs were determined for every solution in the population, the solutions were ranked according to their fitness value. Then new populations (of solutions) were generated via selection, crossover and mutation of individuals (solutions) within the current population. These newer populations (microgrid configurations) in the new generation were more fitter (lower LCOE) than the ones in the previous generation. Such a procedure was repeated until the maximum number of generations was reached. The optimal solution (microgrid configuration) was the fittest individual in the current generation (values used for genetic operators presented in Table 3).

4.3. Exhaustive Search Algorithm (ESA)

Being a stochastic optimization routine, the GA search cannot guarantee an exact optimal solution but can converge to a near optimum solution which is close to the global minimum [47]. Therefore, results from the GA search were verified using a ESA (brute force method) that examined the entire design space to find all possible microgrid configurations able to provide uninterrupted power. The solution which had the lowest value of LCOE was then selected from all possible solutions. This entire process was done only for the single-home case given the computational demand to perform a ESA analysis. For the 10 and 50 home analyses, only the GA search was used.

The ESA is portrayed in Fig. 8. In this procedure, the EMS was implemented for every hour in a typical year for all possible microgrid configurations in the design space. If at any instant for a given solution (microgrid configuration), the LPSP was not zero then that solution was discarded. For all other solutions, LCOE and GHG emissions were computed and recorded.

5. Results and discussions

5.1. Scenario I: Single home analysis

For a single home, the performance of ESA and GA techniques were compared with each other and the results are depicted in Table 4. The upper and lower bound of variables for both the techniques (N_{PV} N_{WT} N_{LB} N_{BDG} N_{FC} N_{EM} N_{H2}) were set at (6 6 1 6 6 6 1) and (0 0 0 0 0 0 0) respectively. The ESA evaluated each of these combinations (total of 67,228) before converging to the optimum (lowest cost) solution, while the GA evaluated only 7% of them. Owing to this difference, the approximate run time for ESA was 60 min while that for GA was 3 min.

Both the ESA and the GA converged to the same set of solutions. The three lowest cost microgrid configurations for all the three cities are shown in Table 5. The optimum configurations for Tucson and Lubbock show a greater preference towards PVs than WTs since the energy potential for solar is higher than that of wind (Fig. 2) in those locations. The converse is true for the case of Dickinson. Because the energy potential of wind is significantly higher than that of solar (Fig. 2), the

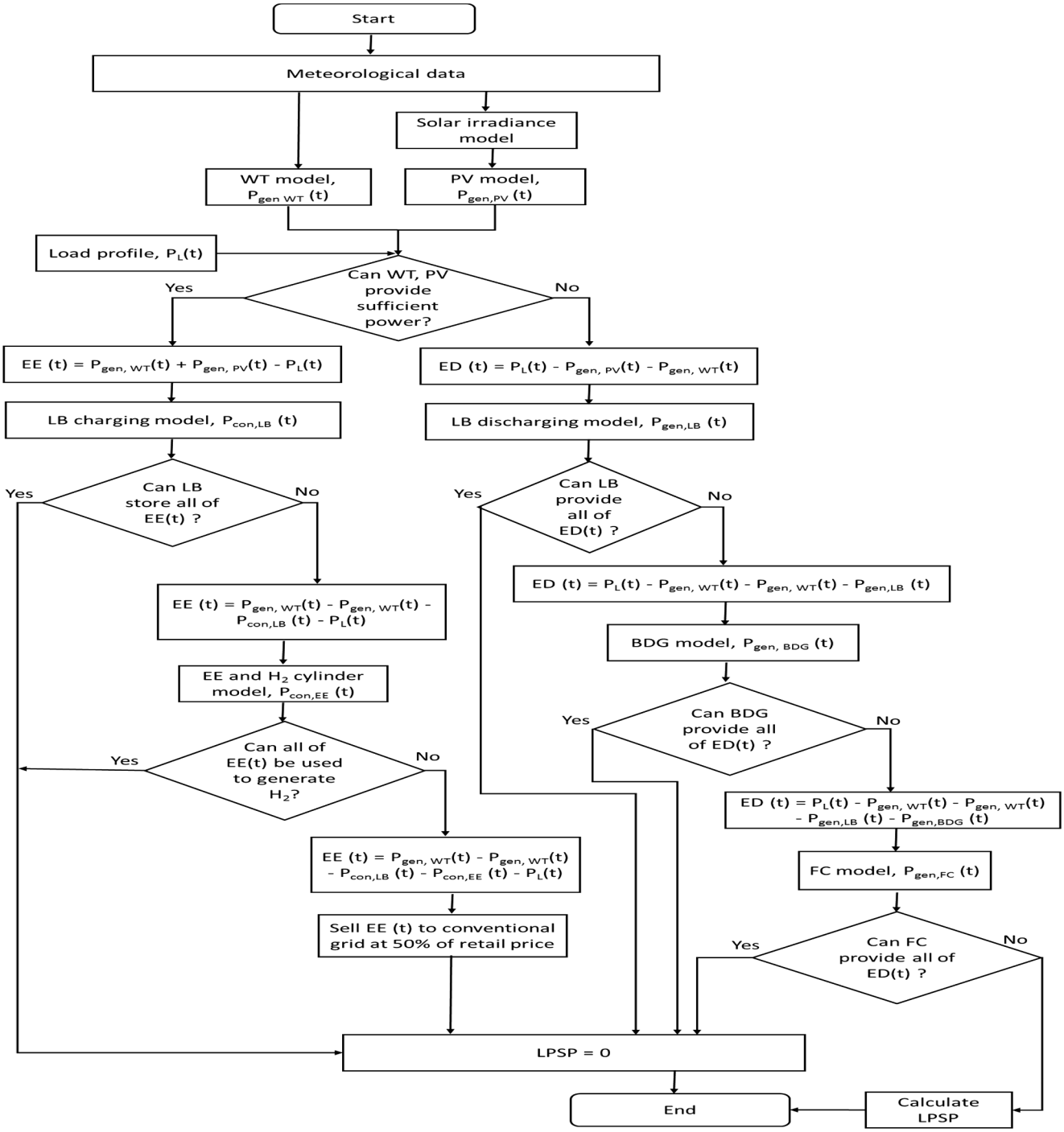


Fig. 6. Energy management strategy (EMS) in a microgrid.

solutions show a greater preference towards WTs than PVs. The configurations in all the cities show a same level of preference to BDGs which are in the range of 3–4 kW. Furthermore, they also show a disinclination towards FCs and EMs due to their high costs.

The analysis of monthly energy generated for the lowest cost microgrid configuration (solution no 1) for Tucson, Lubbock and Dickinson are shown in Figs. 9–11 respectively. As shown in Table 5, the PV generation capacity of Tucson and Lubbock is 5–6 times as that of Dickinson, while their WT generation capacity is approximately half as that of Dickinson. As a result the microgrids in Tucson (Fig. 9) and Lubbock (Fig. 10) were shown to generate about 65% and 35% of their energy via their PV and WT units respectively while in the case of Dickinson (Fig. 11), the corresponding energy generation amounts were 7% and 82% respectively. For Tucson and Lubbock, the energy analyses also reveal that as a result of energy demand increase (from June to September), less energy is sold to the conventional grid and more energy is generated via backup sources (LB, BDG).

The amount of energy generated by individual component cannot be solely predicted from the value of its name plate capacity as it is also dependent on the EMS. For instance in the case of Tucson, even though the installed capacity of WT (3 kW) was lower than BDG (4 kW), the total annual energy generated by WT was 6 times as that generated by BDG. This is because according to the EMS, WT is a primary source of power while BDG is an emergency source. For primary sources of energy (WT and PV), the energy produced is mainly dependent on the installed capacity while in case of emergency sources it is dependent on

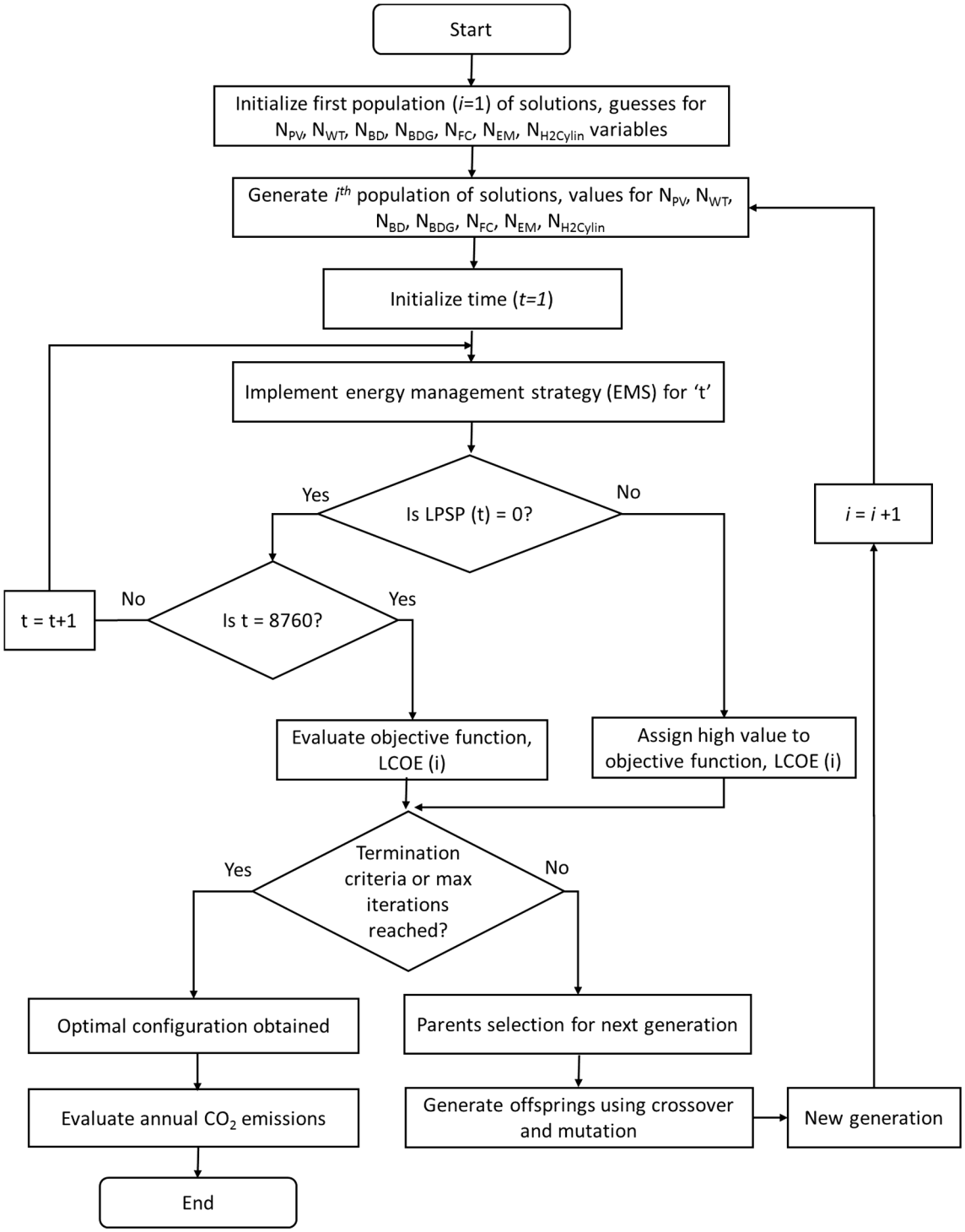


Fig. 7. Genetic algorithm (GA) flowchart.

the difference in the values between energy generated by primary sources and the energy demand (energy deficit, ED).

The LCOE ($/kWh) for individual components in the microgrid (solution no 1) for Tucson, Lubbock and Dickinson are shown in Figs. 12–14 respectively. For Tucson and Lubbock, the LCOE was 14% and 9% higher than that observed in Dickinson which can be attributed to two factors, namely, absence of LBs and high share of energy generation through WTs. LBs' absence caused a reduction in LCOE since the

Table 3
Genetic operators values used for GA [46].

Parameter	Population size	Maximum generations	Crossover probability	Mutation probability	Elitism probability
Value	200	700	0.8	0.01	0.05

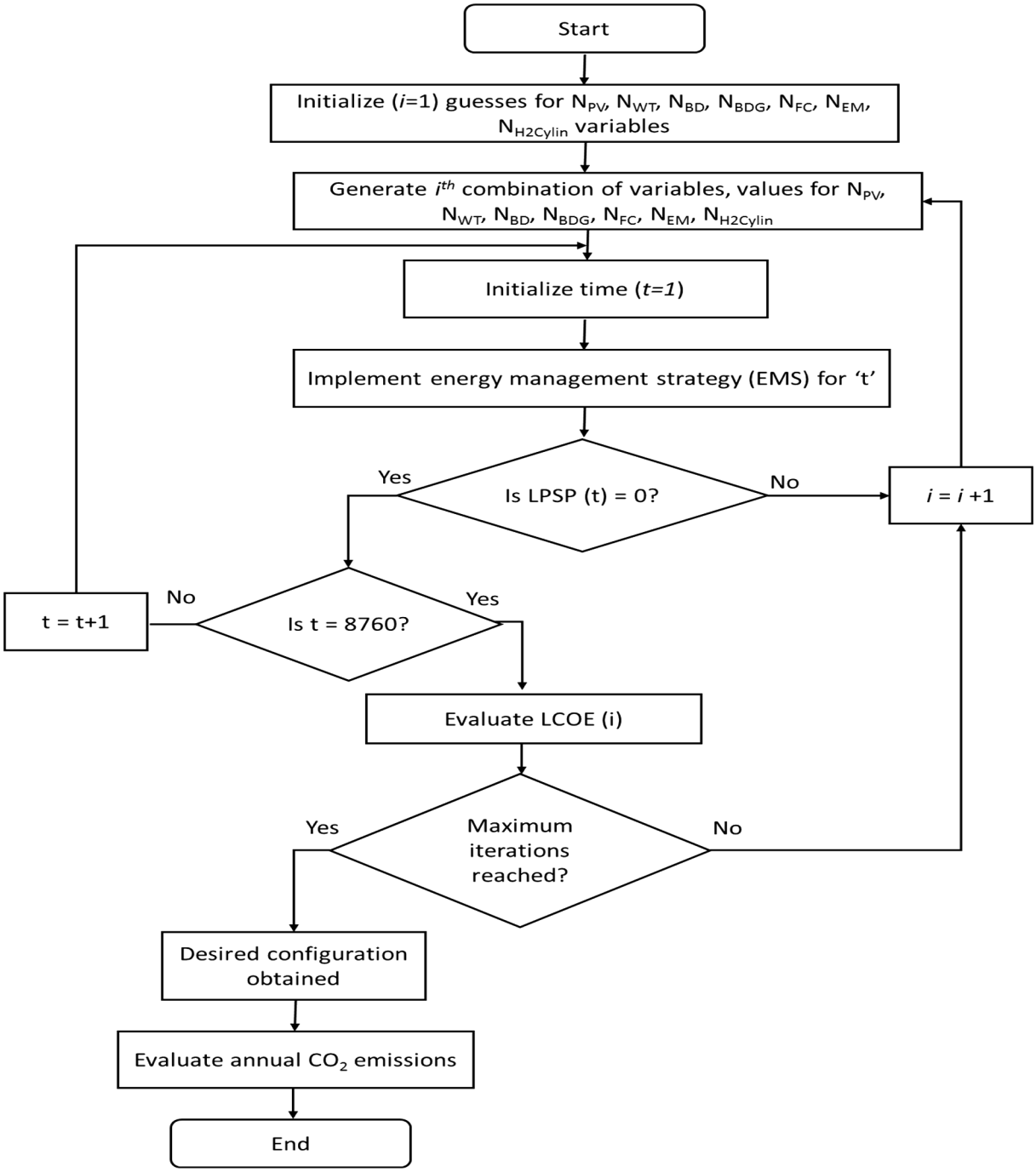


Fig. 8. Flowchart for ESA.

annualized cost of a single unit of battery (22 kWh capacity) is approximately 8 times as that of PV or WT. In addition to this, the share of energy generation (as % of total energy generated) through WT for Dickinson is approximately 2.5 times as that of Tucson or Lubbock. As energy generation through WTs is cheapest method of production available within microgrid (see Figs. 12–14), the LCOE for Dickinson was found to be lowest among all the three cities.

The net LCOE for all the three cities was found to be approximately 4 times as that of the retail rate of electricity in their respective states. Even though the net LCOE of the microgrid is significantly higher than the retail rate of electricity, the LCOE for individual components such as PV or WT may lie below it. For instance in Tucson and Lubbock, the average annual LCOE from WT alone is approximately 50% lower than the retail electricity rates in those locations respectively. Furthermore, for all the three cities, the annual average component LCOEs from PV ($0.071/kWh) and WT ($0.064/kWh) lie at the lower end of cost spectrum, while those from LB ($0.103/kWh) and BDG ($0.122/kWh) lie at the higher end. This is mainly because PV and WT, being the primary sources of energy, produce large amounts of energy while LB and BDG which act as emergency sources produce small amounts of energy (shown in Figs. 9–11). The purchase of biodiesel also contributes to the increase in costs for power generation through BDGs.

The sale of excess electricity to conventional grid does not significantly affect the net LCOE. For Tucson and Lubbock, it brings down the value of LCOE by only 6%, while for Dickinson the drop is 14%. The drop is more for Dickinson because its share of excess energy (as a share of net generated by PV and WT, which is eventually sold) generated is approximately 1.6 times as that of Tucson or Lubbock.

Table 4
Performance analysis of ESA and GA for a single home.

	Tucson, AZ	Lubbock, TX	Dickinson, ND
Maximum number of possible combinations	67,228	67,228	67,228
ESA performance statistics			
Number of combinations evaluated	67,228	67,228	67,228
Number of final set of solutions obtained	25,273	34,679	37,954
Run time (mins)	51	62	67
GA performance statistics			
Number of combinations evaluated	4481	4761	4271
Number of final set of solutions obtained	70	70	70
Number of generations of population	63	67	60
Run time (mins)	3.25	3.09	1.8

For the cities of Tucson, Lubbock and Dickinson, Figs. 15–17 depict the monthly LCA GHG emissions for the individual components of the microgrid along with their comparison with those emitted from a conventional electric grid respectively. For Tucson, Lubbock and Dickinson, the corresponding annual GHG emissions are 7, 8 and 11 times smaller than those emitted by their conventional electric grid equivalents.

The net emissions follow the electricity consumption patterns since they are directly proportional to the amount of energy generated by the microgrid. Additionally, they are also dependent on individual components' emission factors (Table 2). For Tucson and Lubbock, 64% of total annual emissions were attributed to PV component as it produced approximately 65% of their total annual energy. However in the case of Dickinson, even though 82% of total annual energy was generated by WTs, the share of annual emissions attributed to them was just 27%. The majority of emissions were attributed to energy production by BDGs (63%) although their annual share of energy generation was only 11%. This is because the emission factor (kg CO_2 emitted/kWh) of BDG is 17 times as that of WT (see Table 2).

5.2. Scenario II: Ten home analyses

For ten home analyses, the upper bound of solution domain for variables (N_{PV} N_{WT} N_{LB} N_{BDG} N_{FC} N_{EM} N_{H2}) was set at (60 60 10 60 60 60 10) while the lower bound was set at zero. Within this solution space, a simple calculation revealed that the number of possible integer combinations would be 1.02×10^{11}. The ESA technique would approximately take 7.75×10^{7} mins or 147 years to evaluate every combination before determining the lowest cost microgrid configuration. It was shown previously (single home analysis) that the GA converges to the optimum solution within fraction of the time taken by the ESA while also maintaining high levels of accuracy. Therefore, in the case of ten homes only GA was used to find the optimum solution. The performance of the GA was summarized in Table 6.

The top three lowest cost microgrid configurations for the three cities as predicted by the GA are depicted in Table 7. Due to the increase in electricity demand by a factor of ten, the net generation capacity also got expanded within the range of 8.5–10 for all the three cities. The net GHG emissions also increased by a factor of 10, 11 and 8 for Tucson, Lubbock and Dickinson respectively.

The energy distribution patterns for Tucson and Dickinson were similar to those of the single home and therefore are not shown here but are included in the SID. However, these patterns were different for the case of Lubbock and are shown in Fig. 18. For Dickinson, the presence of LBs did not affect these patterns in a major way since only 6% of the total energy generated by PV and WT was stored in them. For Tucson, the share of energy generated by PV rose by 7% while that of WT and BDG fell by 5% and 2% respectively. The share of energy stored in LB also fell by 3%. For Lubbock, the share of energy generated by PV (as a share of total energy generated by PV, WT, BDG) fell by 22% while that of WT and BDG rose by 20% and 2% respectively. Furthermore, the share of energy stored in LB (as a share of total energy generated by PV and WT) also fell by 11%. As less energy was stored in LB, the share of excess energy sold to the grid (as a share of total energy generated by PV and WT) surged by 13%. These results for Lubbock indicate that, as microgrid size increases, greater preference was given to WTs than PVs while converse is true for the case of Tucson.

The LCOE for Lubbock and Dickinson are shown in Figs. 19 and 20 respectively. The cost analysis for Tucson was not shown (but shown in SID) here as it was similar to that of a single home. Relative to single home case, the annual drop in net LCOE for Lubbock and Dickinson was 8% and 12% respectively. For Lubbock, the drop was attributed to 25% and 65% decline in component LCOEs for PV and LB units respectively. For Dickinson, the drop was attributed to 38% decline in LCOE for BDGs. It is interesting to note that the net LCOE drop in Dickinson occurred even after addition of capital intensive components (three LBs) to the microgrid. Despite its drop in costs, the annual LCOE values for Lubbock and Dickinson were still 4.5 and 3.6 times as that of the retail costs respectively.

Figs. 21–23 illustrate the LCA GHG emissions for the lowest cost ten home microgrid configuration in Tucson, Lubbock and Dickinson respectively. Relative to the single home scenario, the net LCA GHG emissions approximately increased by a factor of 10, 11 and 8 times for Tucson, Lubbock and Dickinson respectively. For Tucson and Lubbock, BDGs' share of total emissions increased by 12% and 24% while PVs' share decreased by 8% and 22% respectively. The corresponding emissions' share of LB in these cities also fell by 5% and 7%. In case of Dickinson, the share of PV, WT and LB in total emissions increased by 2%, 5% and 5% respectively at the expense of BDG's share. These results indicate that as microgrid size is increased, the share of individual

Table 5
Lowest cost microgrid configurations for all the three cities as predicted by ESA and GA.

Solution number	N_{PV} (kW)	N_{WT} (kW)	N_{LB} (of 22 kW each)	N_{BDG} (kW)	N_{FC} (kW)	N_{EM} (kW)	N_{H2} (kW)	LCOE ($/kWh)	GHG (kg CO_2 eq./year)
Tucson, AZ									
1	6	3	1	4	0	0	0	0.427	773
2	6	4	1	4	0	0	0	0.432	749
3	5	5	1	4	0	0	0	0.435	690
Lubbock, TX									
1	5	2	1	3	0	0	0	0.407	600
2	5	3	1	3	0	0	0	0.413	586
3	6	1	1	3	0	0	0	0.414	685
Dickinson, ND									
1	1	5	0	4	0	0	0	0.373	683
2	0	6	0	4	0	0	0	0.374	666
3	1	6	0	4	0	0	0	0.374	674

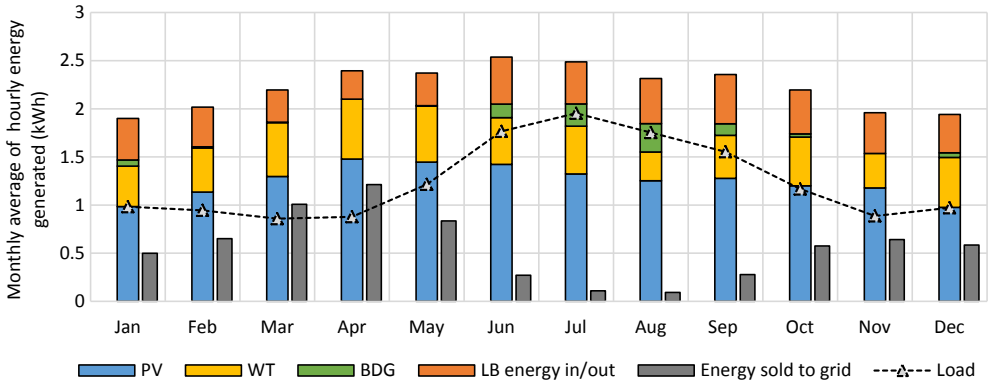


Fig. 9. Analysis of monthly energy generated for the lowest cost microgrid configuration (solution #1) in Tucson, AZ.

components' emissions changes and nature of this change is dependent on microgrid's location. For Tucson, Lubbock and Dickinson, the corresponding annual GHG emissions are 7, 8 and 9 times smaller than those emitted by their conventional electric grid equivalents.

5.3. Scenario III: Fifty home analyses

For fifty home analyses, the upper bound of solution domain for variables (N_{PV} N_{WT} N_{LB} N_{BDG} N_{FC} N_{EM} N_{H2}) was set at (300 300 50 300 300 300 50) while the lower bound was set at zero. Within this solution space, the number of possible integer combinations were found to be 6.42×10^{15}. Based on the single home analyses, the ESA technique would approximately take 4.78×10^{12} mins or 9.09×10^{6} years to evaluate every combination before determining the lowest cost microgrid configuration. It was shown also previously that the GA converges to the optimum solution within fraction of the time taken by the ESA while maintaining high levels of accuracy. Therefore, in the case of fifty homes only GA was used to find the optimum solution. The performance of the GA is summarized in Table 8.

The top three lowest cost microgrid configurations for all the cities as predicted by the GA are depicted in Table 9. As compared to the single home scenario, increasing the electricity demand by a factor of fifty caused the net generation capacity of the microgrid to increase by a similar factor. There was not much difference observed in the LCOEs for ten home and fifty home microgrids. The net GHG emissions increased by a factor of 50, 57 and 38 for Tucson, Lubbock and Dickinson respectively when compared with the single home scenario.

For all the cities, the share of energy generation, emissions and LCOE values of individual microgrid components for lowest cost configuration (solution no. 1) are approximately similar to that of the ten homes scenario. Therefore, these are not shown here but are included in the SID. In case of Lubbock, it is interesting to note the presence of FCs and H_2 tanks in lowest cost microgrid configurations numbered 2 and 3 (Table 9). These results suggest that as microgrid size increases, it might be actually cheaper to generate energy from FCs than from any other sources thereby demonstrating an economies of scale effect.

The LCOEs for all the three scenarios (1, 10 and 50 homes) mentioned above were in the range of $0.32–0.42/kWh. Furthermore, the sale of surplus amount of electricity did not have a major economic impact as it lowered the LCOE by only 7–15%. The LCOEs were approximately 3.5–4.5 times as that of retail cost of electricity in their respective cities. Even though it might not be economically feasible to build a microgrid in these cities, there might be an economic case for cities that have higher electricity retail rates. For instance, in 2017 the average retail electricity price in the state of Hawaii was $0.26/kWh [44]. Microgrids would relatively become more economically competitive and feasible in such states. A detailed economic analysis would be needed to ascertain the economic feasibility.

One of the drawbacks of this work is that it can only conduct the techno-economic and environmental assessment of a small scale microgrid (up to 50 homes). To perform the analyses for larger capacity, the economic costs need to be correspondingly adjusted to those of a larger scale (or utility scale).

5.4. Comparison of economic results with similar microgrid analyses in literature

The economic results obtained in this work are similar to those recently reported in the literature regarding small scale microgrids with the exclusion of any subsidies. The comparison of LCOE (adjusted to 2018$ value) with those available in literature are shown in Table 10. It should be noted that the LCOE for microgrid is calculated on the basis

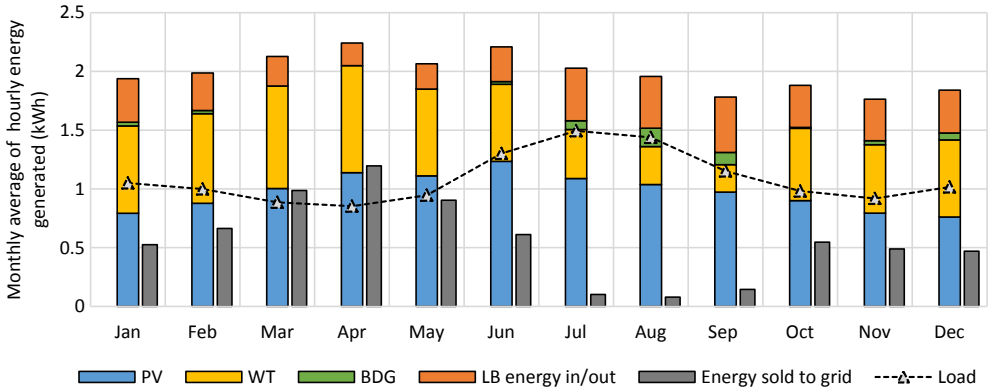


Fig. 10. Analysis of monthly energy generated for the lowest cost microgrid configuration (solution #1) in Lubbock, TX.

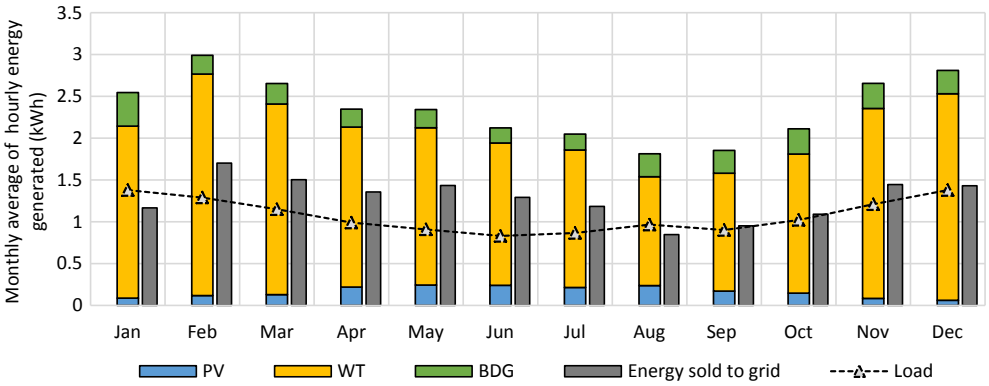


Fig. 11. Analysis of monthly energy generated for the lowest cost microgrid configuration (solution #1) in Dickinson, ND.

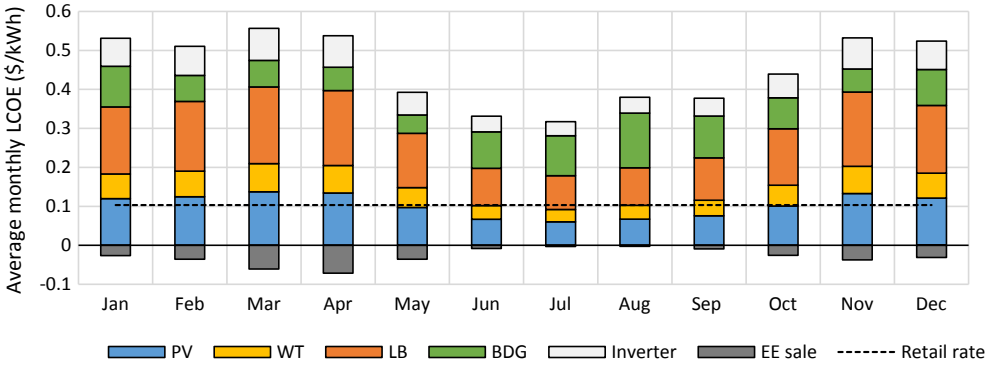


Fig. 12. Component wise average monthly LCOE ($/kWh) for lowest cost microgrid configuration (solution #1) in Tucson, AZ.

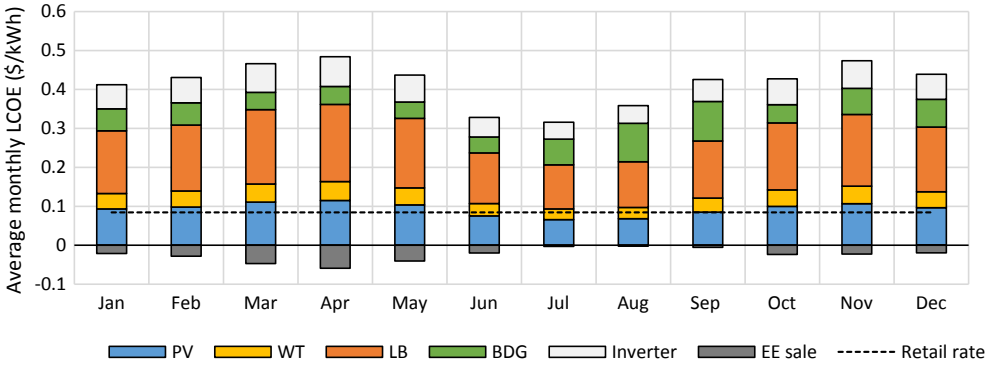


Fig. 13. Component wise average monthly LCOE ($/kWh) for lowest cost microgrid configuration (solution #1) in Lubbock, TX.

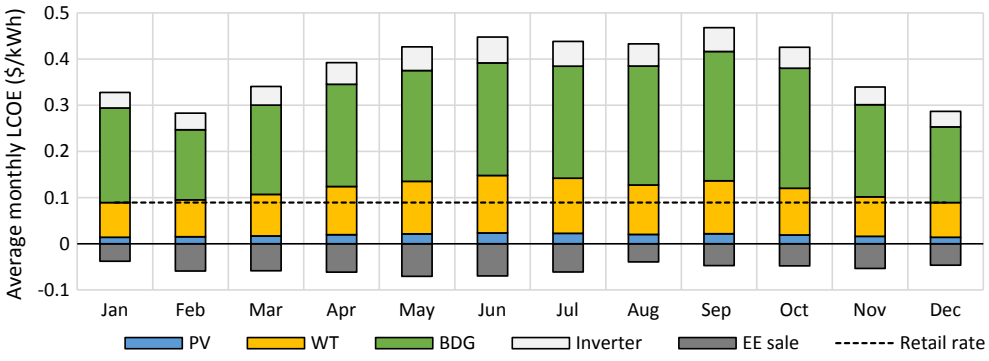


Fig. 14. Component wise average monthly LCOE ($/kWh) for lowest cost microgrid configuration (solution #1) in Dickinson, ND.

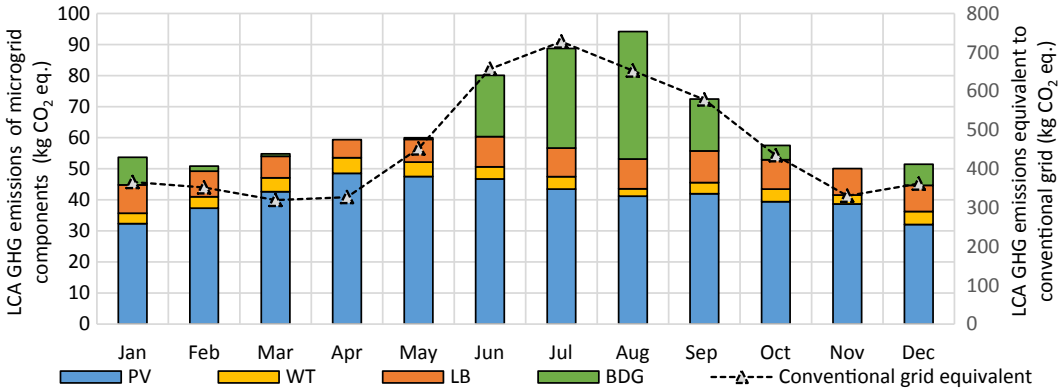


Fig. 15. Monthly LCA GHG emissions (kg CO_2 eq.) for lowest cost single home microgrid configuration (solution #1) in Tucson, AZ.

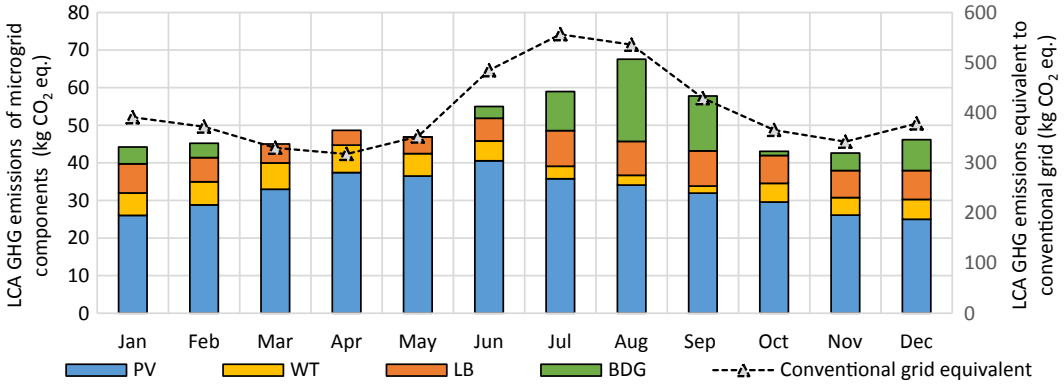


Fig. 16. Monthly LCA GHG emissions (kg CO2 eq.) for lowest cost single home microgrid configuration (solution #1) in Lubbock, TX.

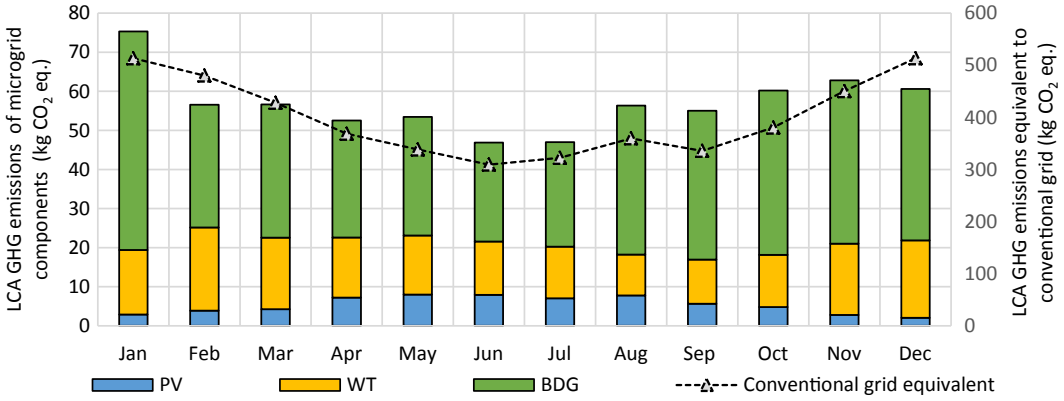


Fig. 17. Monthly LCA GHG emissions (kg CO2 eq.) for lowest cost single home microgrid configuration (solution #1) in Dickinson, ND.

Table 6
Performance analysis of GA for the case of ten homes.

	Tucson, AZ	Lubbock, TX	Dickinson, ND
Maximum number of possible combinations	1.02×10^{11}	1.02×10^{11}	1.02×10^{11}
GA performance statistics			
Number of combinations evaluated	8191	10,501	7001
Number of solutions obtained	70	70	70
Number of generations	116	149	99
Run time (mins)	6.08	6.65	4.6

of local acquisition (capital) costs of the individual components. These costs can significantly differ from country to country. However, this comparison is valuable as it can give an insight regarding the economic feasibility of microgrids in the US and also across the world. Ensuring global economic feasibility is essential in order to tackle the common environmental conundrum of anthropogenic climate change.

6. Conclusions

In this work, a methodology was developed that assessed the technical, economic and environmental performance of a small scale microgrid composed of seven components, including: PV, WT, LB, BDG, FC, EM and H_2 tanks. This novel methodology has been applied to analyze microgrid design in three US cities namely Tucson, Arizona, Lubbock Texas, and Dickenson, North Dakota.

Five main conclusions have been drawn from this study which together inform microgrid design:

Table 7
Lowest cost microgrid configurations for ten homes as predicted by GA.

Solution Number	N_{PV} (kW)	N_{WT} (kW)	N_{LB} (of 22 kW each)	N_{BDG} (kW)	N_{FC} (kW)	N_{EM} (kW)	N_{H2} (kW)	LCOE ($/kWh)	GHG (kg CO_2 eq./year)
Tucson, AZ									
1	54	35	8	39	0	0	0	0.421	7864
2	54	34	8	39	0	0	0	0.421	7902
3	54	36	8	39	0	0	0	0.422	7828
Lubbock, TX									
1	37	35	4	29	0	0	0	0.378	6761
2	38	36	4	29	0	0	0	0.379	6786
3	37	37	4	28	0	0	0	0.387	6714
Dickinson, ND									
1	9	45	3	31	0	0	0	0.329	5269
2	10	45	3	31	0	0	0	0.329	5278
3	8	46	3	31	0	0	0	0.329	5246

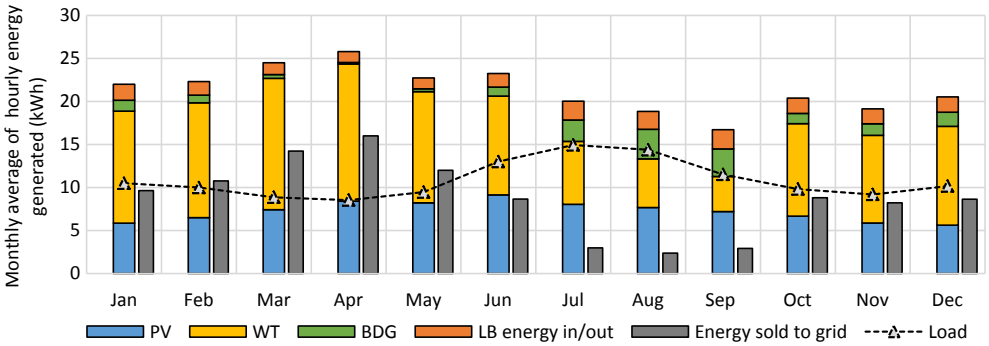


Fig. 18. Analysis of monthly energy generated for lowest cost ten home microgrid configuration (solution #1) in Lubbock, TX.

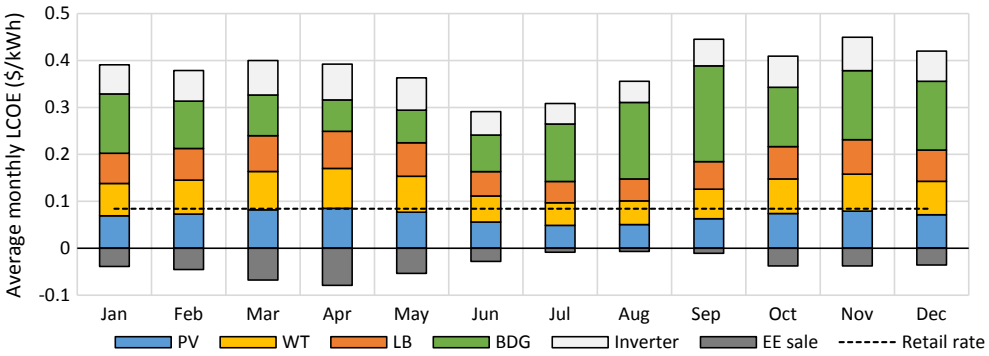


Fig. 19. Average monthly LCOE ($/kWh) for lowest cost ten home microgrid configuration (solution #1) in Lubbock, TX.

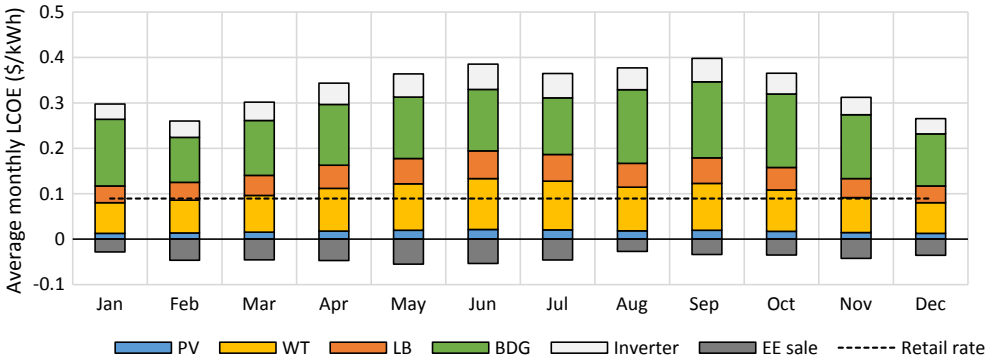


Fig. 20. Average monthly LCOE ($/kWh) for lowest cost ten home microgrid configuration (solution #1) in Dickinson, ND.

1. Based on the regional electricity load profile and availability of renewable resources, an optimum configuration for a standalone microgrid that provided the lowest LCOE was found using a stochastic genetic optimization algorithm (GA). A brute force method (ESA) was also used to evaluate the LCOE for all possible microgrid designs. Comparing the optimum microgrid design found with the ESA

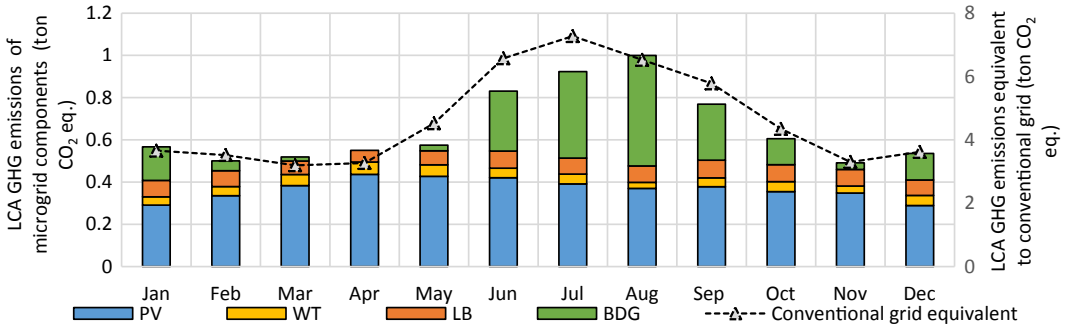


Fig. 21. Monthly LCA GHG emissions (ton CO_2 eq.) for lowest cost ten home microgrid configuration (solution #1) in Tucson, AZ.

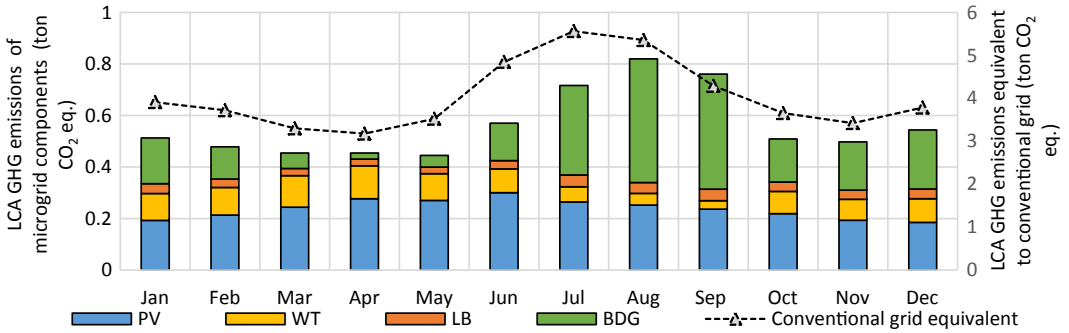


Fig. 22. Monthly LCA GHG emissions (ton CO_2 eq.) for lowest cost ten home microgrid configuration (solution #1) in Lubbock, TX.

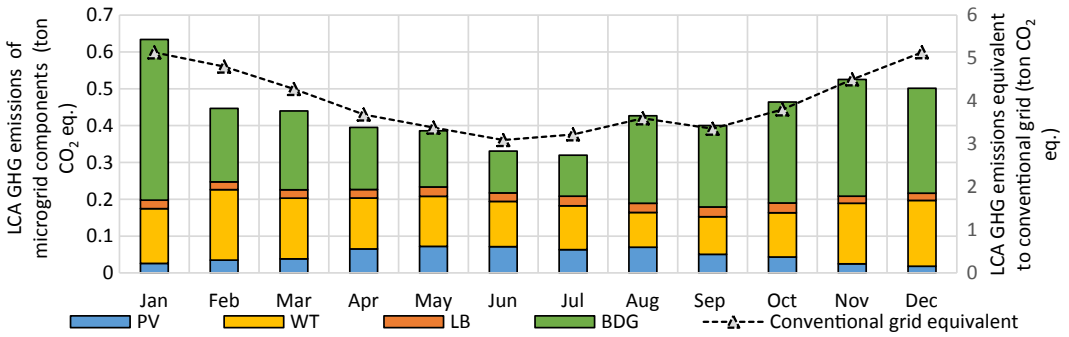


Fig. 23. Monthly LCA GHG emissions (ton CO_2 eq.) for lowest cost ten home microgrid configuration (solution #1) in Dickinson, ND.

Table 8
Performance analysis of GA for the case of fifty homes.

	Tucson, AZ	Lubbock, TX	Dickinson, ND
Maximum number of possible combinations	6.42×10^{15}	6.42×10^{15}	6.42×10^{15}
GA performance statistics			
Number of combinations evaluated	10,991	20,651	10,781
Number of solutions obtained	70	70	70
Number of generations	156	294	153
Run time (mins)	7.7	14.73	6.66

method to that found using the GA method confirmed the ability of the GA method to identify the optimum design using significantly less computational resources.
2. Scaling up the microgrid size (1–10 homes) reduced the LCOE by 7–12% thereby demonstrating the economies of scale effect.
3. Neglecting financial subsidies, the LCOEs were in the range of $0.32–0.42/kWh and were approximately 3.5–4.5 times as that of their local retail electricity cost.
4. The average individual component LCOE values for lowest cost microgrid configurations for PV ($0.066/kWh) and WT ($0.07/kWh) were found to lie at the lower end of cost spectrum while those for LB ($0.0866/kWh) and BDG ($0.119/kWh) were found to lie at the higher end.
5. The environmental footprint of microgrid was found to be extremely low as they were found to be approximately 1/10th as those emitted by an equivalent conventional electric grid.

Future work should be carried out to determine the economic feasibility for a utility scale microgrid that incorporates the social cost ($/ton) of carbon which is a measure of long term economic damage done by CO_2 emissions. Such an assessment that incorporates the technical, economic and environmental aspects would serve as a firm basis that would aid lawmakers in shaping energy and environmental policies of the future.

Table 9
Lowest cost microgrid configurations for fifty homes as predicted by GA.

Solution Number	N_{PV} (kW)	N_{WT} (kW)	N_{LB} (of 22 kW each)	N_{BDG} (kW)	N_{FC} (kW)	N_{EM} (kW)	N_{H2} (kW)	LCOE ($/kWh)	GHG (kg CO_2 eq./year)
Tucson, AZ									
1	277	172	43	192	0	0	0	0.419	38,760
2	278	173	43	192	0	0	0	0.419	38,763
3	277	173	43	192	0	0	0	0.419	38,725
Lubbock, TX									
1	185	188	18	143	0	0	0	0.377	34,432
2	195	253	18	148	3	0	1	0.393	34,234
3	190	191	18	166	2	0	1	0.395	34,636
Dickinson, ND									
1	49	213	17	152	0	0	0	0.326	25,812
2	49	214	17	152	0	0	0	0.326	25,792
3	48	214	17	152	0	0	0	0.326	25,784

Table 10
Comparison of LCOE (adjusted to 2018$) with other studies available in the literature focusing on small scale microgrids.

Sr No.	Microgrid components	Location	LCOE ($/kWh)	Reference
1.	PV-WT-LB-BDG-FC-H_2	US	0.32–0.42	This study
2.	PV	US	0.13–0.16	[48]
4.	PV-WT-DG	Singapore	0.19–0.30	[6]
5.	PV-WT-DG-LB	Saudi Arabia	0.05–0.08	[9]
6.	PV-WT-DG-LB	India	0.27–0.3	[10]
8.	PV-WT-FC	Iran	0.55–0.81	[8]

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Conflict of interest

The authors declared that there is no conflict of interest.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enconman.2018.11.072.

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